

2023 HKDSE Physics

Paper 1B — Structured Questions

Original questions and official answers with detailed English reasoning. Calculations show the governing relationships, substitution and units; each structured subpart is presented separately.

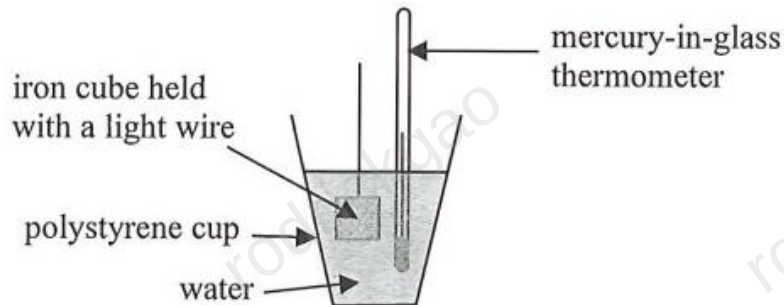
Editable questions with original diagrams, official answers and detailed English reasoning. Each subpart is separated for clarity.

Structured Questions

Official answers are placed directly below each question.

Question 1

An iron cube of mass 200 g is heated to a few hundred degrees Celsius (say at T °C). It is then quickly transferred into a polystyrene cup containing 600 g of water. The cube is totally immersed in the water as shown in Figure 1.1. The temperature of the heated iron cube can be estimated in this experiment.



Assume that no heat is lost to the surroundings and neglect the heat capacity of the polystyrene cup.

- (a) The temperature of water recorded by the thermometer rises from 25 °C to 33 °C. Estimate T .
Given: The specific heat capacities of iron and water are $450 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ and $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$ respectively. (3 marks)
- (b) What would happen to the water when this iron cube of high temperature just touches the water surface? (1 mark)
- (c) Explain why the actual temperature of the iron cube is higher than that estimated in (a). (2 marks)
- (d) (i) A mercury-in-glass thermometer is unsuitable for measuring the temperature of the heated iron cube directly. Why? (1 mark)
- (ii) Suggest a kind of thermometer that can be used to measure the temperature of the heated iron cube directly. (1 mark)

Official marking scheme

Paper 1 Section B

Solution	Marks	Remarks
1. (a) Heat given out by iron cube = Heat gained by water $0.2 \times 450 \times (T - 33) = 0.6 \times 4200 \times (33 - 25)$ $T = 257\text{ }(^{\circ}\text{C})$	2M 1A	
	3	
(b) Some water might become steam (immediately).	1A	
Or Some water in the cup might splash.	1A	
	1	
(c) It is because some heat is lost to the surroundings through vaporization of water / as latent heat.	1A 1A	
Or It is because some heat is lost to the surroundings (or polystyrene cup) / The water might not be well stirred, otherwise, the water should have reached a temperature higher than $33\text{ }^{\circ}\text{C}$ (i.e. underestimated).	1A 1A	
	2	
(d) (i) Poor thermal contact between the glass bob of the thermometer and the iron cube.	1A	
	1	
(ii) Infra-red (IR) thermometer.	1A	
	1	

(a) With no heat loss, heat lost by iron equals heat gained by water: $0.200(450)(T-33) = 0.600(4200)(33-25)$. Thus $T-33 = 224\text{ }^{\circ}\text{C}$ and $T = 257\text{ }^{\circ}\text{C}$. Use the common final temperature $33\text{ }^{\circ}\text{C}$ for both materials.

(b) Some water may vaporize immediately into steam; splashing may also occur. The hot cube can heat a small amount of water locally above its boiling point.

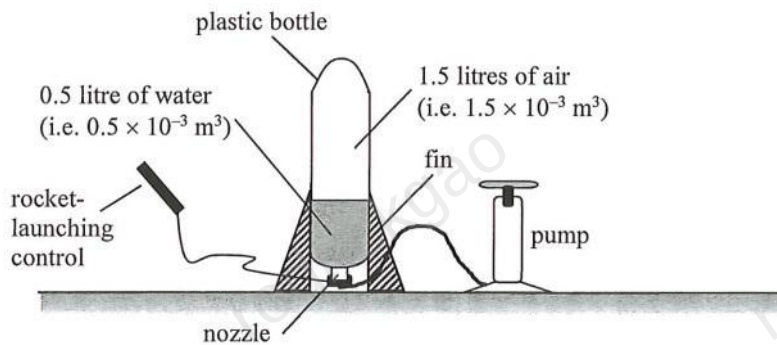
(c) Some energy from the cube is carried away as latent heat by escaping water vapour or transferred to the surroundings. This energy is not counted in the measured warming of the remaining water, so part (a) underestimates the cube's initial temperature.

(d)(i) The thermometer's glass bulb makes poor thermal contact with the solid cube, preventing a reliable direct temperature measurement.

(d)(ii) Use an infrared thermometer suitable for the temperature range.

Question 2

Figure 2.1 shows a water rocket made from a 2-litre plastic bottle fitted with fins. The bottle is filled with water such that 1.5 litres of air is trapped inside.



Initially, the trapped air is at 27°C and atmospheric pressure. Given: atmospheric pressure = $1.0 \times 10^5 \text{ Pa}$; density of air = 1.20 kg m^{-3} .

(a) Find the mass m_0 of the air initially trapped inside the bottle. (2 marks)

Air is pumped slowly into the bottle until the total mass of air inside becomes $4m_0$. Assume that the temperature remains at 27°C (300 K) and the volume of the trapped air remains constant.

(b) (i) Find the pressure of the trapped air. (2 marks)

(ii) With reference to the kinetic theory, explain the increase in pressure. (2 marks)

(iii) The area of the water surface is 0.014 m^2 . Estimate the increase in force acting on the water due to the trapped air. (2 marks)

(c) The mass of the plastic bottle can be neglected.

(i) After the control is pulled, a jet of water is ejected from the nozzle. Explain, in terms of Newton's laws, why the rocket moves upwards. (2 marks)

(ii) Just after pulling the control, what is most of the energy stored in the pressurized air converted to? (1 mark)

Official marking scheme

Solution	Marks	Remarks
2. (a) $m_0 = \text{density} \times \text{volume}$ $= 1.20 \times (1.5 \times 10^{-3})$ $= 1.8 \times 10^{-3} \text{ kg or } 1.8 \text{ g}$	1M 1A	
	2	
(b) (i) $p_1 V = n_1 R T$ and $p_2 V = n_2 R T$ As $m_0 \rightarrow 4m_0$, $n_2 = 4n_1$ $\therefore p_2 = \frac{n_2 R T}{V} = \frac{4n_1 R T}{V} = 4p_1 = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$	1M 1A	
Or $pV = nRT$ $p \propto n$ (as V , R and T are constant) Since $m_0 \rightarrow 4m_0$, thus $n \rightarrow 4n$, p becomes 4 times the original value ($= 1.0 \text{ atm}$), $p = 4.0 \times 10^5 \text{ Pa (or } 4.0 \text{ atm)}$	1M 1A	
	2	
(ii) More air molecules hit unit area of surface of wall or bottle / water surface (as $n \rightarrow 4n$ moles of molecules trapped in the same volume), The surface of wall or bottle / water surface is hit more frequently OR rate of change of total momentum, i.e. force, per unit area of wall or bottle / water surface increases, pressure increases.	1A 1A	
	2	
(iii) $\Delta F = \text{increase in pressure} \times \text{area}$ $= [(4 - 1) \times 10^5] \times 0.014$ $= 4200 \text{ N}$	1M 1A	error-carried-forward (e.c.f.) from (b)(i)
	2	
(c) (i) The pressure force acting on the water results in its change of momentum (a water jet). Magnitude of the force = rate of change of momentum. According to Newton's third law of motion, an equal but opposite force / reaction / thrust acts on the rocket. (According to Newton's second law of motion,) the reaction force causes the rocket to accelerate / move upwards.	1A 1A	
	2	
(ii) kinetic energy of (the jet of) water / work done on (the jet of) water	1A	
	1	

(a) Convert 1.5 litres to $1.5 \times 10^{-3} \text{ m}^3$. The initial mass is $m_0 = \rho V = 1.20(1.5 \times 10^{-3}) = 1.8 \times 10^{-3} \text{ kg} = 1.8 \text{ g}$.

(b)(i) From $pV = nRT$, p is proportional to n at fixed volume and temperature. Quadrupling the mass quadruples the number of moles, so $p = 4(1.0 \times 10^5) = 4.0 \times 10^5 \text{ Pa}$, an absolute pressure.

(b)(ii) The unchanged temperature means unchanged average molecular kinetic energy. There are four times as many molecules in the same volume, so collisions with unit wall area occur more frequently. The momentum transferred per unit time per unit area increases, giving greater pressure.

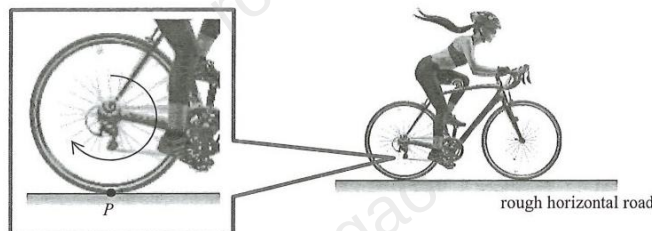
(b)(iii) The increase is $\Delta F = \Delta p A = (4.0 \times 10^5 - 1.0 \times 10^5)(0.014) = 4200 \text{ N}$. Use the pressure increase, not the final pressure alone.

(c)(i) Pressure accelerates the water downwards, changing its momentum. By Newton's third law, an equal and opposite thrust acts upwards on the rocket. The resultant upward force accelerates the rocket according to Newton's second law.

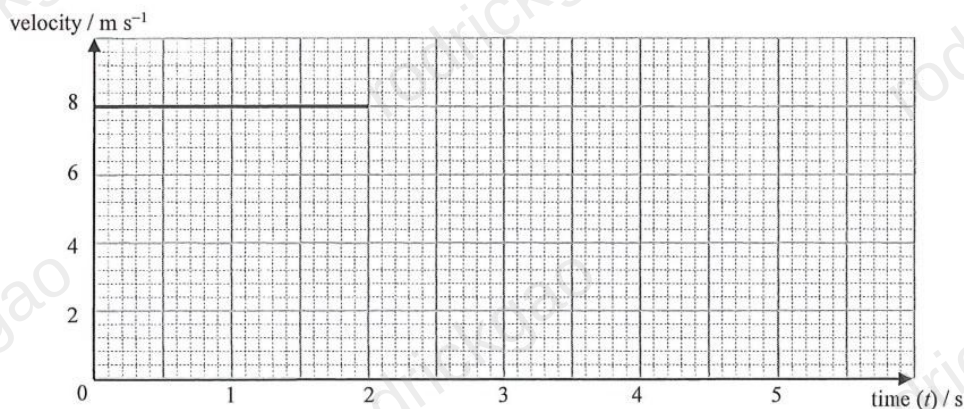
(c)(ii) Most of the stored energy is initially converted into kinetic energy of the ejected water jet.

Question 3

A cyclist is riding a bicycle at a constant speed $v = 8.0 \text{ m s}^{-1}$ on a rough horizontal road as shown in Figure 3.1.



- The rear wheel rotates clockwise without slipping. P is the point of contact between the rear wheel and the road. Indicate the direction of the frictional force acting on the rear wheel at P. (1 mark)
- The effective resistive force acting on the cyclist and bicycle is 17.0 N. Estimate the mechanical power developed by the cyclist. (2 marks)
- At $t = 2.0 \text{ s}$, the cyclist sees an obstacle 9 m in front and applies the brakes as soon as possible. His reaction time is 0.2 s. The bicycle then decelerates uniformly and stops at $t = 3.8 \text{ s}$.
- Complete the velocity–time graph for the motion from $t = 2.0 \text{ s}$ to $t = 3.8 \text{ s}$. (2 marks)



- The total mass of the cyclist and bicycle is 65 kg. Calculate the resultant resistive force that produces the deceleration. (2 marks)
- Find the distance between the bicycle and the obstacle when the bicycle stops. (3 marks)
- Figure 3.2 shows a knee pad containing a padded cushion. Explain the physical principle by which the cushion reduces injury in a collision. (2 marks)

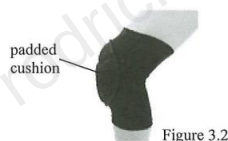
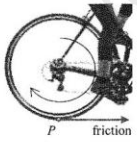
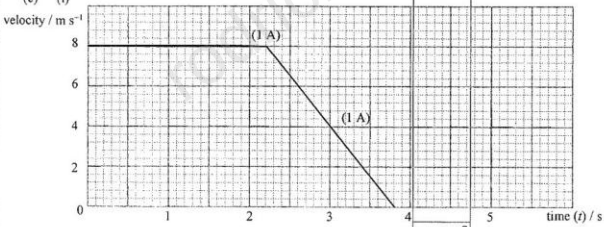


Figure 3.2

Official marking scheme

Solution	Marks	Remarks
3. (a) 	1A	
(b) Mechanical power delivered (for work done against the resistive force) $P = Fv$ $= 17.0 \times 8.0$ $= 136 \text{ W}$	1M 1A	
(c) (i) 	2	
(ii) Deceleration: $\frac{0-8}{3.8-2.2} = -5.0 \text{ m s}^{-2}$ Resistive force $= 65 \times 5.0$ $= 325 \text{ N}$	1M 1A	e.c.f. from (c)(i)
Alternative method: Braking distance $= \frac{1}{2}(8)(3.8 - 2.2) = 6.4 \text{ m}$ $Fs = \frac{1}{2}mv^2 - 0$ $F(6.4) = \frac{1}{2}(65)(8)^2$ $F = 325 \text{ N}$	1M 1A	
(iii) Total distance travelled after $t = 2.0 \text{ s}$ $= \text{area under the graph from } t = 2.0 \text{ s to } t = 3.8 \text{ s}$ $= \frac{0.2+1.8}{2} \times 8.0 = 8.0 \text{ m}$ Distance between the bicycle and the obstacle when the bicycle stops $= 9 - 8$ $= 1 \text{ m}$	1M/1A 1M 1A	
(d) The padded cushion can increase the contact time of the knee and the ground such that a smaller force acts on the knee.	1A 1A	

(a) Draw friction at P horizontally to the right. The driven rear wheel tends to push the road backwards, so the road exerts forward static friction on the wheel.

(b) At constant speed the driving force balances resistance. Mechanical power $P = Fv = 17.0(8.0) = 136 \text{ W}$.

(c)(i) Continue the horizontal line at 8.0 m s^{-1} from $t = 2.0 \text{ s}$ to $t = 2.2 \text{ s}$ for the reaction time. Then draw a straight line down to zero at $t = 3.8 \text{ s}$.

(c)(ii) Braking lasts $3.8 - 2.2 = 1.6 \text{ s}$. Thus $a = (0 - 8.0)/1.6 = -5.0 \text{ m s}^{-2}$ and the resultant resistive force is $65(5.0) = 325 \text{ N}$, opposite the motion.

(c)(iii) Reaction distance $= 8.0(0.2) = 1.6 \text{ m}$. Braking distance $= \frac{1}{2}(8.0)(1.6) = 6.4 \text{ m}$. Total stopping distance is 8.0 m , leaving $9.0 - 8.0 = 1.0 \text{ m}$ before the obstacle.

(d) The cushion deforms and increases the stopping time of the knee. For the same momentum change, average force $F = \Delta p / \Delta t$ is smaller, reducing injury.

Question 4

- (a) A hollow cylindrical drum rotates uniformly about a vertical axis. A block appears pinned to the inner wall and does not fall, as shown in Figure 4.1(a). Figure 4.1(b) shows a side view of the block at the wall.

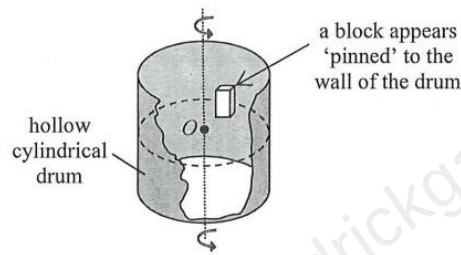


Figure 4.1 (a)

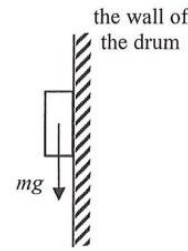
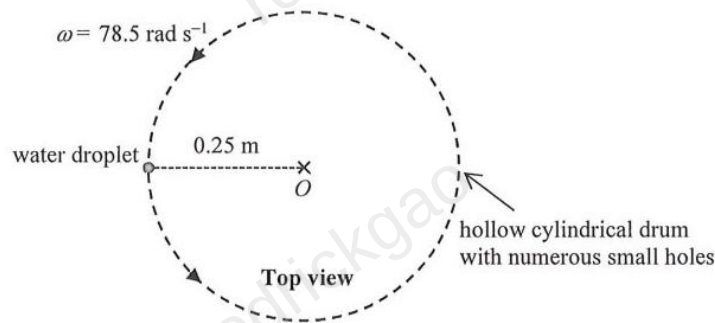


Figure 4.1 (b)

- (i) Besides its weight mg , two forces act on the block. Draw and label them in Figure 4.1(b). (2 marks)
- (ii) Which force provides the centripetal force on the block? (1 mark)
- (b) A washing machine has a hollow drum with holes in its wall. During spinning, the drum rotates at angular speed $\omega = 78.5 \text{ rad s}^{-1}$.
- (i) Find the rotation speed in revolutions per minute. (2 marks)

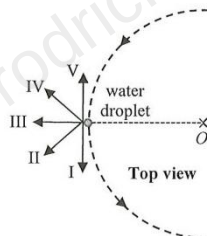
The drum has radius 0.25 m. Figure 4.2 is a top view of a water droplet rotating anticlockwise with the drum.



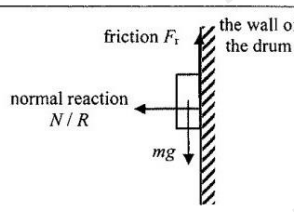
- (ii) Calculate the centripetal force on a water droplet of mass $2 \times 10^{-5} \text{ kg}$. (2 marks)

Suppose the droplet is just about to escape through a hole in the drum.

- (iii) Estimate the speed of the droplet. (2 marks)
- (iv) Which of the directions I–V shown represents the direction of motion of the droplet just after it escapes? (1 mark)



Official marking scheme

Solution		Marks	Remarks
4. (a) (i)		2A	
		2	
(ii)	Normal reaction (force acting on drum/wall of the drum by the object)	1A	
		1	
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ <u>Or</u> $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s and}$ $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M	
		1A	
		2	
(ii)	centripetal force $= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$	1M	
		1A	
		2	
(iii)	$v = r\omega$ $= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M	
		1A	
		2	
(iv)	1	1A	
		1	

(a)(i) Draw static friction vertically upwards along the wall and the normal reaction horizontally to the left, towards the drum's axis. Friction balances mg vertically.

(a)(ii) The inward normal force exerted by the drum wall on the block supplies its centripetal force. The equal opposite force on the wall acts on a different object.

(b)(i) One revolution is 2π radians, so rotation speed = $[78.5/(2\pi)]60 = 749.6 \approx 750 \text{ rev min}^{-1}$.

(b)(ii) $F_c = m\omega^2 r = (2 \times 10^{-5})(78.5)^2(0.25) = 0.0308 \text{ N}$, towards the axis.

(b)(iii) The instantaneous tangential speed is $v = r\omega = 0.25(78.5) = 19.6 \text{ m s}^{-1}$.

(b)(iv) Direction I. Just after escape the droplet retains its instantaneous tangential velocity; it does not suddenly acquire an outward radial velocity.

Question 5

A sound wave from a loudspeaker propagates to the right. Figure 5.1(a) shows the equilibrium positions of air particles *a* to *u*, and Figure 5.1(b) shows their positions at $t = 0$.

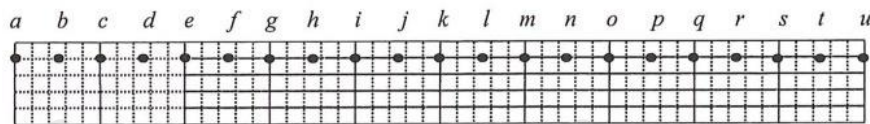


Figure 5.1(a)

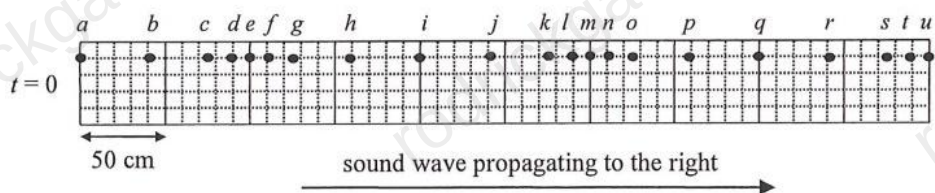
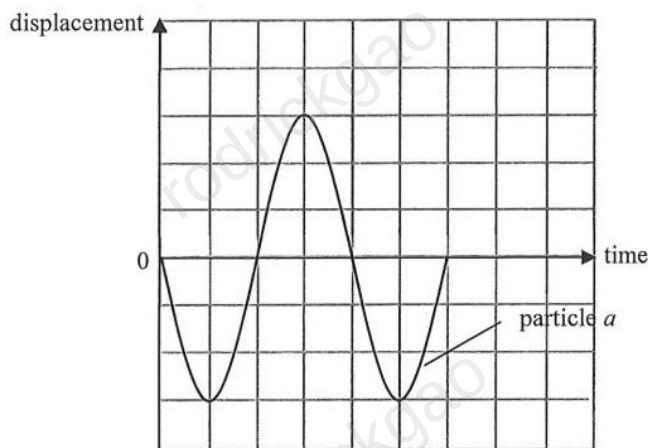


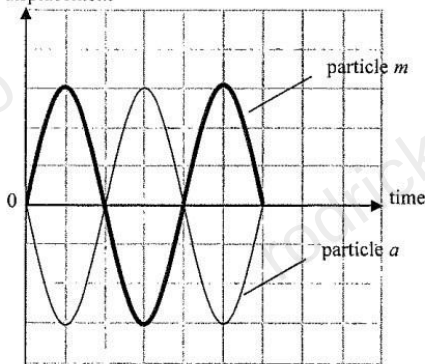
Figure 5.1(b)

- (a) At $t = 0$, state the particle that is
- at the centre of a rarefaction; (1 mark)
 - displaced furthest to the right from its equilibrium position. (1 mark)
- (b) Each grid spacing represents 50 cm. The speed of sound is 340 m s^{-1} .
- Find the wavelength of the sound wave. (1 mark)
 - Find the frequency of the sound wave. (2 marks)
- (iii) Figure 5.2 shows the displacement–time graph of particle *a*. Sketch on the same axes the displacement–time graph of particle *m*. (2 marks)



- (c) How would the position of particle *h* at $t = 0$ change if the sound were louder? (1 mark)

Official marking scheme

Solution	Marks	Remarks
5. (a) (i) $a / i / q$	1A	
	1	
(ii) $c / k / s$	1A	
	1	
(b) (i) wavelength (4 grids) = $4 \times 50 \text{ cm}$ = 200 cm or 2 m	1A	
	1	
(ii) speed = frequency $\times$ wavelength frequency = speed $\div$ wavelength = $340 \div 2$ = 170 Hz	1M 1A	
	2	
(iii)	2A	
displacement		
		
time		
	2	
(c) Displacement increases, i.e. further to the left.	1A	
	1	

(a)(i) Particle a, i or q is at a rarefaction centre, where neighbouring particles are most spread out. Any one is sufficient.

(a)(ii) Particle c, k or s has the greatest displacement to the right relative to its own equilibrium position. Any one is sufficient.

(b)(i) Adjacent rarefaction centres are four labelled 50 cm grid widths apart, so $\lambda = 4(0.50) = 2.0 \text{ m}$.

(b)(ii) From $v = f\lambda$, $f = 340/2.0 = 170 \text{ Hz}$.

(b)(iii) Particles a and m are separated by 1.5 wavelengths at equilibrium, so they are in antiphase. Draw a sinusoid with the same amplitude and period as a, reflected about the time axis: m begins at zero displacement and initially moves positive.

(c) Louder sound has greater displacement amplitude. At this instant h is displaced leftwards, so it moves further to the left of its equilibrium position.

Question 6

Rainbows are sometimes seen after a rainfall (Figure 6.1(a)). Tiny water droplets in the air each act like a prism to disperse sunlight into a spectrum of colours. The most common rainbow that we see is called a primary rainbow, in which white light from the Sun undergoes reflection once within each water droplet. Under suitable conditions, a double rainbow with an additional secondary rainbow formed above the primary rainbow can be seen (Figure 6.1(b)).

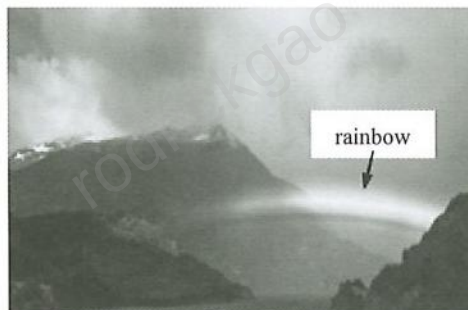


Figure 6.1(a)

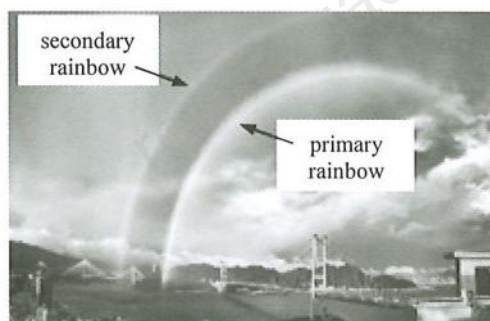
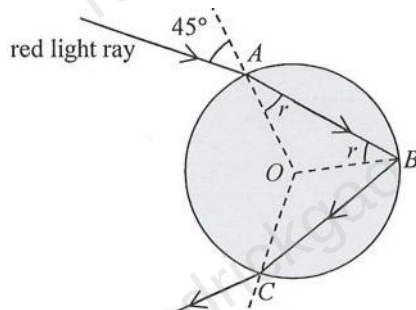


Figure 6.1(b)

Assume that the water droplets are spherical. The refractive index of water for red light is 1.325.

(a) Figure 6.2 shows a red light ray entering a water droplet.

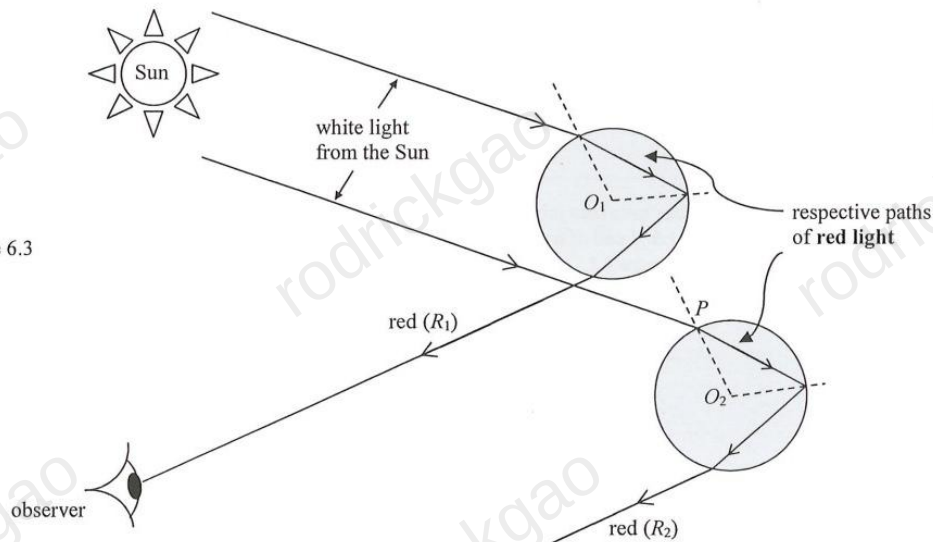


(i) The angle of incidence at A is 45° . Find the angle of refraction r . (2 marks)

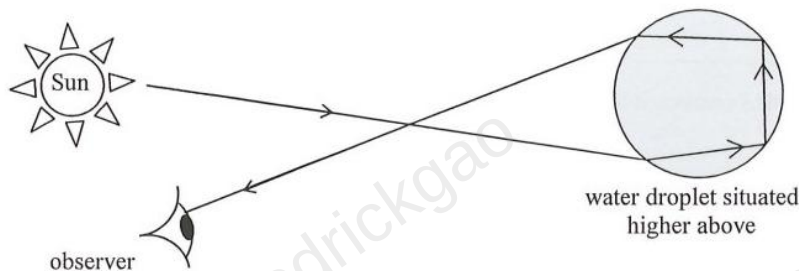
(ii) Calculate the critical angle c at a water–air boundary and justify whether the reflection at B is total internal reflection. (3 marks)

- (b) (i) Figure 6.3 shows an observer viewing a rainbow. O_1 and O_2 are the centres of an upper and a lower water droplet. The paths R_1 and R_2 of red light are drawn; R_1 can be seen by the observer. The lower droplet enables a violet ray to reach the observer. Sketch the path of this violet ray from P. The refractive index of water for violet light is slightly greater than that for red light. (2 marks)

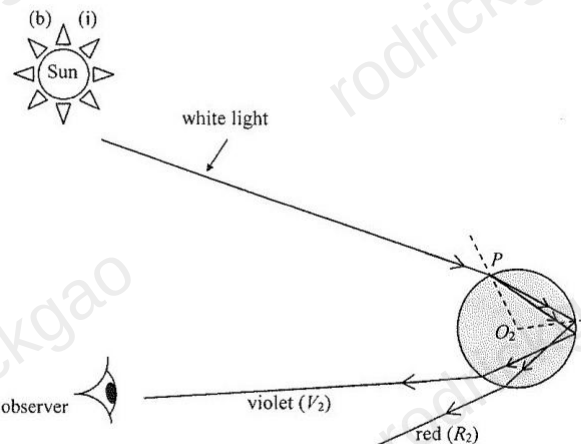
Figure 6.3



- (ii) A secondary rainbow is formed by light reflected twice inside droplets situated higher above the ground, as shown in Figure 6.4. Explain why it is dimmer than the primary rainbow. (2 marks)



Official marking scheme

Solution	Marks	Remarks
6. (a) (i) $\sin r = \frac{\sin 45^\circ}{1.325}$ $r = 32.253454^\circ \approx 32.3^\circ$	1M 1A	
	2	
(ii) $c = \sin^{-1} \frac{1}{1.325}$ $c = 49.000646^\circ \approx 49.0^\circ$ r (or $\angle OBA$) $= 32.3^\circ < c = 49.0^\circ$, therefore total internal reflection does not occur at point B.	1M 1A 1A	
	3	
(b) (i) 	2A	
	2	
(ii) There is more light energy absorbed by water droplet due to a longer path.	1A 1A	
Or There is more light energy transmitted to air at the point of reflection as the light undergoes reflection twice.	1A 1A	
	2	

(a)(i) Snell's law gives $\sin 45^\circ = 1.325 \sin r$. Hence $r = \sin^{-1}(\sin 45^\circ / 1.325) = 32.3^\circ$.

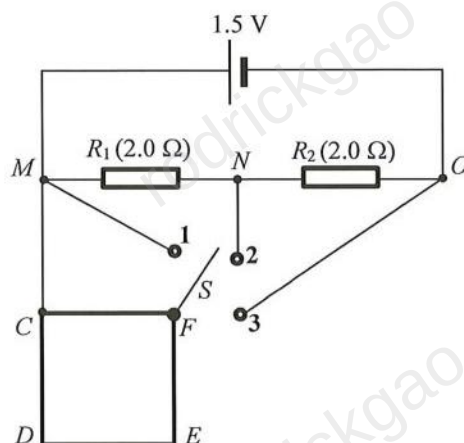
(a)(ii) At the water–air boundary, $\sin c = 1/1.325$, so $c = 49.0^\circ$. The radii OA and OB form an isosceles triangle, giving incidence at B equal to $r = 32.3^\circ$. Since $r < c$, the reflection at B is partial, not total internal reflection.

(b)(i) From P draw the violet ray bending more towards the inward normal than the red ray. Reflect it once at the droplet's far boundary, then refract it out towards the observer, following the violet path in the marking diagram. Its larger refractive index produces the greater overall deviation.

(b)(ii) At each internal reflection some light is transmitted out of the droplet. The additional reflection therefore leaves less energy in the secondary-rainbow ray. Greater absorption along its longer path is another accepted explanation.

Question 7

- (a) A wire of length 0.20 m and cross-sectional area $8.0 \times 10^{-7} \text{ m}^2$ has resistance $0.50 \, \Omega$. Find the resistivity of the material in $\Omega \text{ m}$. (2 marks)
- (b) Four such wires form the square CDEF. They are connected to a 1.5 V cell of negligible internal resistance and two identical $2.0 \, \Omega$ resistors R_1 and R_2 as shown in Figure 7.1.



- (i) To which terminal, 1, 2 or 3, should S be connected so that the current in side CF is greatest? (1 mark)
- (ii) When S is connected to terminal 2, find the equivalent resistance across M and N. (2 marks)
- (iii) With S connected to terminal 2, compare the potential difference across R_2 with that across R_1 . (1 mark)
- (iv) When S is connected to terminal 1, determine the power dissipated by the square coil. Explain your reasoning. (2 marks)

Official marking scheme

Solution	Marks	Remarks
7. (a) $R = \rho \frac{l}{A} \Rightarrow 0.50 = \rho \frac{0.20}{8.0 \times 10^{-7}}$ $\rho = 2.0 \times 10^{-6} (\Omega \text{ m})$	1M 1A 2	
(b) (i) Terminal 3	1A 1	
(ii) When switch S is connected to terminal 2, the equivalent resistance across MN $R = \frac{1}{\frac{1}{2.0} + \frac{1}{0.5} + \frac{1}{0.5+0.5+0.5}}$ $= 0.315789 \, \Omega \approx 0.316 \, \Omega$	1M 1A 2	
(iii) (p.d. across R_2 is) greater than (p.d. across R_1)	1A 1	
(iv) zero as no p.d. across CF / no current in the coil.	1A 1A 2	

(a) From $R = \rho l/A$, $\rho = RA/l = 0.50(8.0 \times 10^{-7})/0.20 = 2.0 \times 10^{-6} \Omega \text{ m}$.

(b)(i) Terminal 3 connects CF directly across the ideal 1.5 V cell, giving the greatest p.d. and therefore the greatest current through CF.

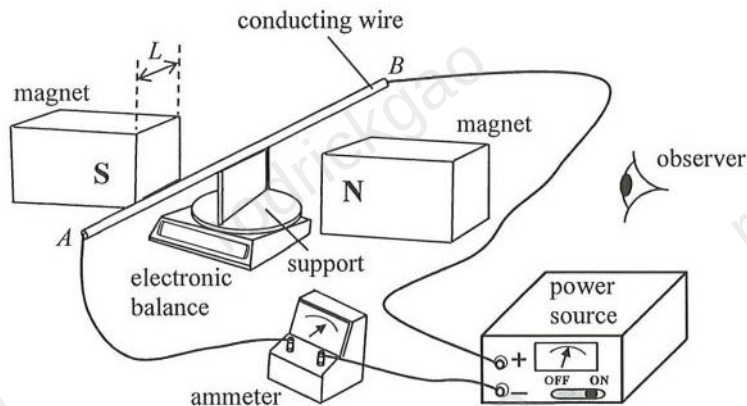
(b)(ii) With terminal 2 selected, three paths connect M and N: $R_1 = 2.0 \Omega$, side CF = 0.50Ω , and CDEF = 1.50Ω . Hence $R_{MN} = [1/2.0 + 1/0.50 + 1/1.50]^{-1} = 0.316 \Omega$.

(b)(iii) The p.d. across R_2 is greater. R_2 carries the sum of the currents through R_1 and the coil; since R_1 and R_2 have equal resistance, $V = IR$ gives the larger p.d. across R_2 .

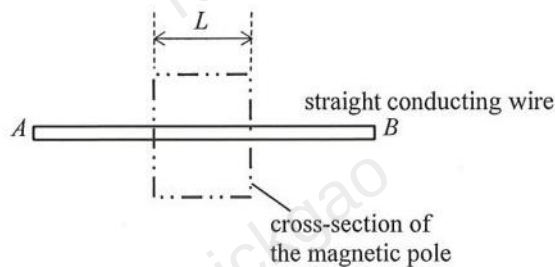
(b)(iv) Power is zero. Terminal 1 connects both C and F to M, so there is no p.d. across either path through the coil, no current, and no electrical energy dissipated in it.

Question 8

A student investigates the force on a straight current-carrying wire between the poles of a magnet. The magnetic field is uniform and perpendicular to the wire. The current is supplied by an adjustable power source. The wire is held horizontally by an insulated support on an electronic balance so that the variation of magnetic force can be registered.



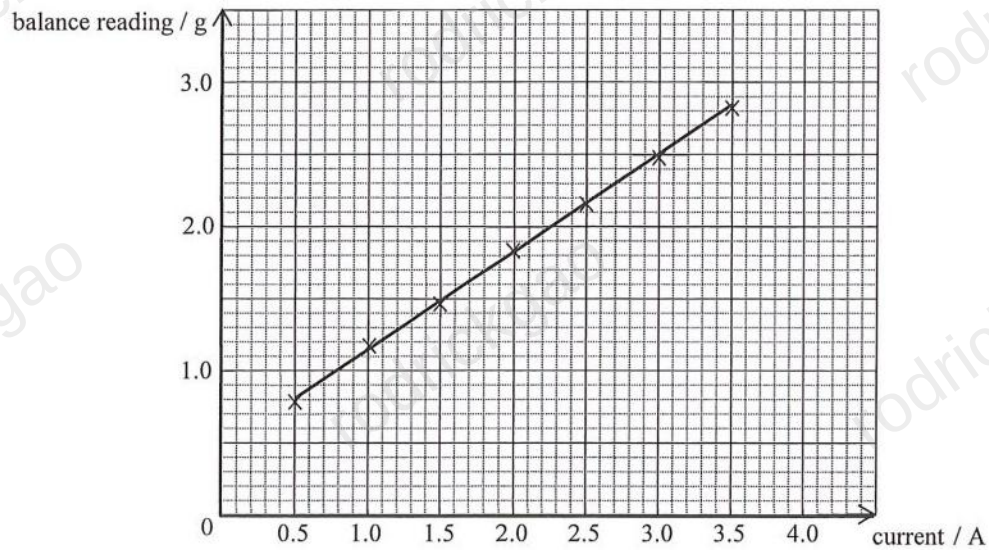
- (a) Mark the + and – terminals of the ammeter in Figure 8.1. (1 mark)
- (b) Figure 8.2 is a side view of AB. Draw and label the directions of the magnetic field, current I and magnetic force F_B . (3 marks)



- (c) Complete the table by stating whether the magnetic force F_B increases, decreases or remains unchanged. (3 marks)

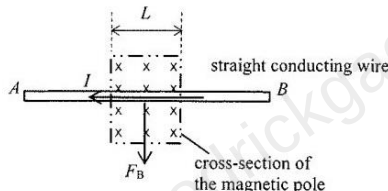
Change	Effect on F_B
Use a stronger magnet of the same dimensions	
Use another support that lowers the wire slightly	
Use a longer wire while keeping the current unchanged	

(d) The graph shows the balance reading against current obtained in the experiment.



- (i) State the procedures used to obtain the data for the graph. (2 marks)
- (ii) Describe how the balance reading varies with current. (1 mark)
- (iii) Explain why the graph does not pass through the origin when the current is zero. (1 mark)

Official marking scheme

Solution		Marks	Remarks							
8. (a)		1A								
		1								
(b)		3A								
		3								
(c)	<table border="1"> <thead> <tr> <th></th> <th>magnetic force F_B</th> </tr> </thead> <tbody> <tr> <td>using a stronger magnet of the same dimensions</td> <td>increase</td> </tr> <tr> <td>using another support such that the position of the horizontal wire is slightly lowered</td> <td>unchanged</td> </tr> <tr> <td>using a longer wire and the current is kept unchanged</td> <td>unchanged</td> </tr> </tbody> </table>			magnetic force F_B	using a stronger magnet of the same dimensions	increase	using another support such that the position of the horizontal wire is slightly lowered	unchanged	using a longer wire and the current is kept unchanged	unchanged
	magnetic force F_B									
using a stronger magnet of the same dimensions	increase									
using another support such that the position of the horizontal wire is slightly lowered	unchanged									
using a longer wire and the current is kept unchanged	unchanged									
		1A								
		1A								
		3								
(d) (i)	(Switch on the power source.) Adjust the power source to deliver a small current, say, 0.5 A to the wire and record the ammeter and the balance readings. Adjust the power source to increase the current stepwise (0.5 A) and record the corresponding balance reading.	1A								
		1A								
		2								
(ii)	The balance reading increases linearly with the current.	1A								
		1								
(iii)	The graph does not pass through the origin because the balance has not been reset to zero initially before the experiment.	1A	Accept: because of the weight of the support / wire							
		1								

(a) Mark the ammeter terminal connected to A positive and the terminal connected to the power source's negative terminal negative. In the drawn meter these are the left and right terminals respectively.

(b) In the side view, draw magnetic-field crosses into the page, current from B to A (leftwards), and magnetic force vertically downwards. These directions satisfy $F = I L \times B$.

(c) A stronger magnet: force increases because B increases.

(c) Lowering the wire slightly: force remains unchanged while it stays in the same uniform field.

(c) A longer wire: force remains unchanged because only the same length L between the pole faces experiences the field; $F_B = BIL$.

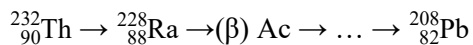
(d)(i) Set a small current and record the ammeter and balance readings. Increase the current in steps and record the corresponding balance reading each time, keeping the wire position and magnet unchanged.

(d)(ii) The balance reading increases linearly with current. It is not directly proportional to current because the intercept is non-zero.

(d)(iii) The balance was not zeroed with the wire and support in place; their weight gives a non-zero reading even when the magnetic force is zero.

Question 9

Part of the decay series of thorium-232 is shown below. Its half-life is 1.41×10^{10} years and the end product lead-208 is stable.



- (a) (i) State the kind of radiation emitted by thorium-232. (1 mark)
- (ii) Determine the numbers of α -decays and β -decays before lead-208 is formed. (2 marks)
- (b) Find the proportion of thorium-232 nuclei that decay in 10 years. Give your answer to 2 significant figures. (2 marks)
- (c) Thorium oxide was formerly added to glass to make thoriated glass with a higher refractive index. Such glass was used in camera lenses manufactured before 1970.



- (i) Although thoriated glass is radioactive, it is practically safe to use a camera fitted with the lens shown. Explain why. (1 mark)
- (ii) Thorium oxide eventually changes to lead oxide, which is black. Explain why there is no need to worry that the lens will turn black after several years. (1 mark)

Official marking scheme

Solution		Marks	Remarks
9. (a) (i)	alpha particle / α	1A	
		1	
	(ii) 6 α -decays, 4 β -decays	1A	
	$4 n_{\alpha} = 232 - 208$	1A	
	$2 n_{\alpha} + (-1) n_{\beta} = 90 - 82$	1A	
	$n_{\alpha} = 6$	2	
	$n_{\beta} = 4$		
	(b) $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/t_{1/2}} = \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$	1M	
	Portion = $1 - \frac{N}{N_0} = 1 - \left(\frac{1}{2}\right)^{\frac{10}{1.41 \times 10^{10}}}$	1A	
	$= 4.9159 \times 10^{-10} \approx 4.9 \times 10^{-10}$ (2 sig. fig.)		
(c) (i)	Aluminium body blocks alpha and/or beta particles emitted from thoriated glass.	1A	$N = N_0 e^{-\lambda t}$ and $\lambda = \frac{\ln 2}{t_{1/2}}$ $\lambda = \frac{\ln 2}{t_{1/2}} = 4.9159 \times 10^{-11} \text{ yr}^{-1}$ $1 - \frac{N}{N_0} = 1 - e^{-\frac{\ln 2}{t_{1/2}} t}$ $= 1 - e^{-\frac{\ln 2}{1.41 \times 10^{10}} (10)}$ $\approx 4.9 \times 10^{-10}$
	Or		
	Short range of α particles	1A	
		1	
	(ii) Thorium-232 has an extremely long half-life.	1A	
	Or		
	As part (b) suggests, only an extremely small fraction (4.9×10^{-10}) of thorium decaying in 10 years.	1A	
		1	

(a)(i) Alpha radiation. The mass number decreases by 4 and the atomic number by 2, identifying emission of an alpha particle.

(a)(ii) The mass-number change requires $n_{\alpha} = (232 - 208)/4 = 6$. After six alpha decays the atomic number would be $90 - 12 = 78$; four beta-minus decays raise it to 82. Thus there are 6 alpha and 4 beta-minus decays.

(b) The surviving fraction is $N/N_0 = \exp[-(\ln 2)t/T]$. The decayed fraction is $1 - \exp[-(\ln 2)(10)/(1.41 \times 10^{10})] \approx (\ln 2)(10)/(1.41 \times 10^{10}) = 4.9 \times 10^{-10}$ (2 s.f.). This is a dimensionless proportion, not an activity or a percentage.

(c)(i) In the illustrated arrangement the aluminium camera body blocks the short-range alpha particles (and beta particles from the decay chain), greatly limiting exposure outside the camera.

(c)(ii) The extremely long half-life means only a minute fraction decays over several years. Too little lead oxide is produced to make the lens noticeably black.